

Nikhilesh Prabhakar

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PhD Candidate. US Citizen - No sponsorship required.

EDUCATION

University of Texas at Dallas

Aug 2020 - Present

Ph.D. and M.S. in Computer Science. Advisor: Dr. Sriraam Natarajan (StARLing Lab). Expected defense: Spring 2027. Jonsson School Graduate Fellowship.

Best Poster Award - Foundational AI, UT Dallas ECS AI Days, 2026 (*Neurosymbolic Imitation Learning with Human Guidance*).

Vellore Institute of Technology

Jul 2016 - May 2020

B.Tech. in Computer Science. CGPA 9.23/10.

RESEARCH INTERESTS

Primary: Neuro-symbolic AI, Automated Planning, Reinforcement Learning, Agentic AI.

Secondary: Multi-agent Systems, Interpretable Policy Learning, Sample-efficient Learning, Sequential Decision-Making.

PUBLICATIONS

N. Prabhakar, V. Balaji, A. Karanam, K. Kersting, S. Natarajan. Neurosymbolic Imitation Learning with Human Guidance: A Privileged Information Approach. *Under Review, ECML, 2026.*

N. Prabhakar, R. Singh, H. Kokel, S. Natarajan, P. Tadepalli. Combining Planning and Reinforcement Learning for Solving Relational Multiagent Domains. *AAMAS, 2025.*

H. Kokel, **N. Prabhakar**, B. Ravindran. Hybrid Deep RePREL: Integrating Relational Planning and RL. *FUSION, 2023.*

SELECTED PROJECTS

Hierarchical-AutoToS: Multi-agent Planning via LLM Code Synthesis

Apr 2026 - Present

Extends AutoToS (AAAI 2025) to multi-agent planning: an LLM produces the entire planning stack as executable code (decomposer, per-subtask successor, completion tests, goal predicate), validated by automated unit-test feedback. Hierarchical decomposition recovers single-agent accuracy on warehouse multi-robot delivery; monolithic joint planning fails.

Automated Predicate Discovery for Relational RL

Apr 2026 - Present

Discovers relational symbolic abstractions from RL experience without hand-written domain knowledge. Reward-adjacent trajectory segmentation, breakpoint-context clustering, and per-operator state abstractions yield zero-shot generalization to novel object counts. Validated on relational Taxi from reward signal alone.

Neurosymbolic Imitation Learning with Privileged Information

Sep 2025 - Feb 2026

Neurosymbolic framework treating expert gaze as privileged information for sample-efficient imitation. Neural perception grounds observations into probabilistic first-order atoms; saliency reweights atom valuations before a differentiable forward-chaining reasoner learns rule-weighted policies. 2.5x the unmodulated baseline; 20x behavior cloning at 10% of training data.

PROFESSIONAL EXPERIENCE

StARLing Lab, University of Texas at Dallas

Aug 2021 - Present

Graduate Research Assistant, Richardson, TX

- Developed neuro-symbolic and relational reinforcement-learning frameworks for hierarchical decision-making and multi-agent coordination; published at AAMAS, AAAI, and FUSION. Co-developed research with collaborators at Oregon State University and TU Darmstadt.

Workflow Chair, AAAI-24 (Vancouver, 2024). Coordinated paper-review workflows for the main technical track and AISI/SRAI special tracks. **ML Consultant, Careband** (Summer 2021). Activity recognition from accelerometer data.

SKILLS & TEACHING

Programming: Python, C++, PDDL, ASP (Clingo), SQL, Java.

Research Methods: RL, Hierarchical/Relational Planning, Neuro-symbolic Reasoning, ILP, Imitation Learning.

Teaching: TA at UT Dallas (ML, Digital Logic), 2021-23; TA at VIT (Intro ML), 2019.